

Task 1: Create a backup of your Music folder (or any folder with at least 3 files) using the tar command and compress it with gzip, saving the output as music_backup.tar.gz in your home directory.

Ans :-

Create the backup

```
tar -czvf ~/music_backup.tar.gz ~/Music
```

If your Music folder does not exist, replace ~/Music with any folder containing at least 3 files, for example:

```
tar -czvf ~/music_backup.tar.gz ~/Documents
```

Verify the backup was created

```
ls -lh ~/music_backup.tar.gz
```

View the archive contents

```
tar -tzf ~/music_backup.tar.gz
```

Task 2: Extract the contents of a tar.gz file (for example, music_backup.tar.gz) into a new directory called restored_music and verify that all files are restored correctly.

Ans :-

Create the restore directory

```
mkdir -p ~/restored_music
```

Extract the archive

```
tar -xzf ~/music_backup.tar.gz -C ~/restored_music
```

Verify the restored files

```
ls -R ~/restored_music
```

Or compare with the original:

```
diff -r ~/Music ~/restored_music/root/Music
```

Note: If the archive was created using ~/Music, it may restore under root/Music or your username depending on how the archive was created.

Task 3: Write a shell command that creates a compressed backup of all .jpg images from your Downloads folder into a file named

images_backup.tar.gz, excluding any files larger than 5MB.
Hint: Use the find command with size filtering and tar with --files-from.

Ans :-

Create a compressed archive of all JPG files in **Downloads** excluding files larger than **5 MB**.

```
find ~/Downloads -type f -name "*.jpg" -size -5M | tar -czvf ~/images_backup.tar.gz --files-from=-
```

Verify the archive

```
tar -tzf ~/images_backup.tar.gz
```

Task 4: Simulate a restore scenario: delete one file from your original Music folder, then use your music_backup.tar.gz archive to restore only the deleted file without overwriting the rest.

Ans :-

Step 1: Delete a file (simulation)

Example:

```
rm ~/Music/song1.mp3
```

Verify:

```
ls ~/Music
```

Step 2: Find the file inside the archive

```
tar -tzf ~/music_backup.tar.gz
```

Example output:

```
root/Music/song1.mp3
```

```
root/Music/song2.mp3
```

Step 3: Restore only the deleted file

```
tar -xzvf ~/music_backup.tar.gz root/Music/song1.mp3 -C ~
```

Or if the archive stores relative paths:

```
tar -xzvf ~/music_backup.tar.gz Music/song1.mp3 -C ~
```

Step 4: Verify restoration

ls ~/Music

You should see the deleted file restored without affecting the other files.